

Law Office Management System

UML and Software Design Project

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Project Overview:

This document presents a UML and software design project for a Law Office Management System. The project includes functional and non-functional requirements, structural and behavioral UML models, CRC cards, and CRUD analysis.

The purpose of this project is to demonstrate software engineering documentation, requirements analysis, object-oriented design, UML modeling, and large-scale software design thinking.

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Introduction

Scenario:

The proposed software is a Law Office Management System that allows people to find lawyers online. The system supports lawyers and clients who have cases registered in court. It provides facilities for lawyers to store and manage previous case details, client information, and upcoming judgment dates. When a case is referred to again, the lawyer can review previous case details and plan for future judgments.

Primary Actors:

- * Lawyer**
- * Client**
- * Visitor**
- *Admin**

Functional Requirements

- 1. Admin login**
- 2. Lawyer registration by Admin**
- 3. Updating lawyer information by Admin**
- 4. Removing a lawyer from the system by Admin**
- 5. Lawyer login**
- 6. Client registration by Lawyer**
- 7. Updating client information by the lawyer**
- 8. Removing a client from the system by the lawyer**
- 9. Adding client case details by Lawyer**
- 10.Updating client case details by the Lawyer**
- 11.Removing selected client case details by Lawyer**
- 12.Client login**
- 13.Viewing and saving client case details by Client**
- 14.Visitor registration by a new user**
- 15.Searching for lawyers by registered visitors**

Non-Functional Requirements

Performance:

90% of system responses should be completed within 2 seconds, except for downloading case details.

Security:

All client case details and personal information must be protected against unauthorized or malicious access.

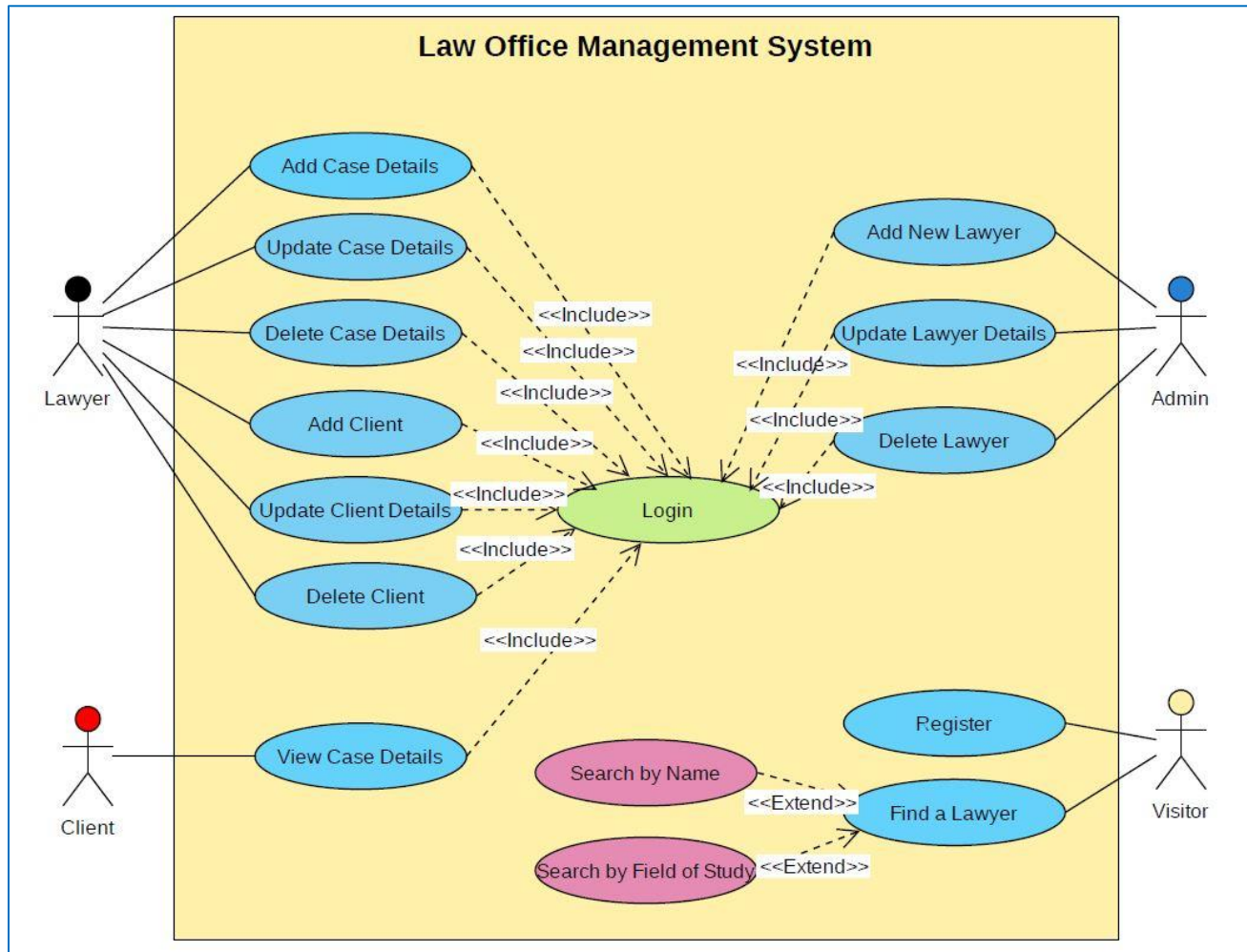
Reliability:

Case data should not be lost due to system failure.

Maintainability:

The system should be maintainable and should support future modifications and the addition of new features.

Use-Case Diagram



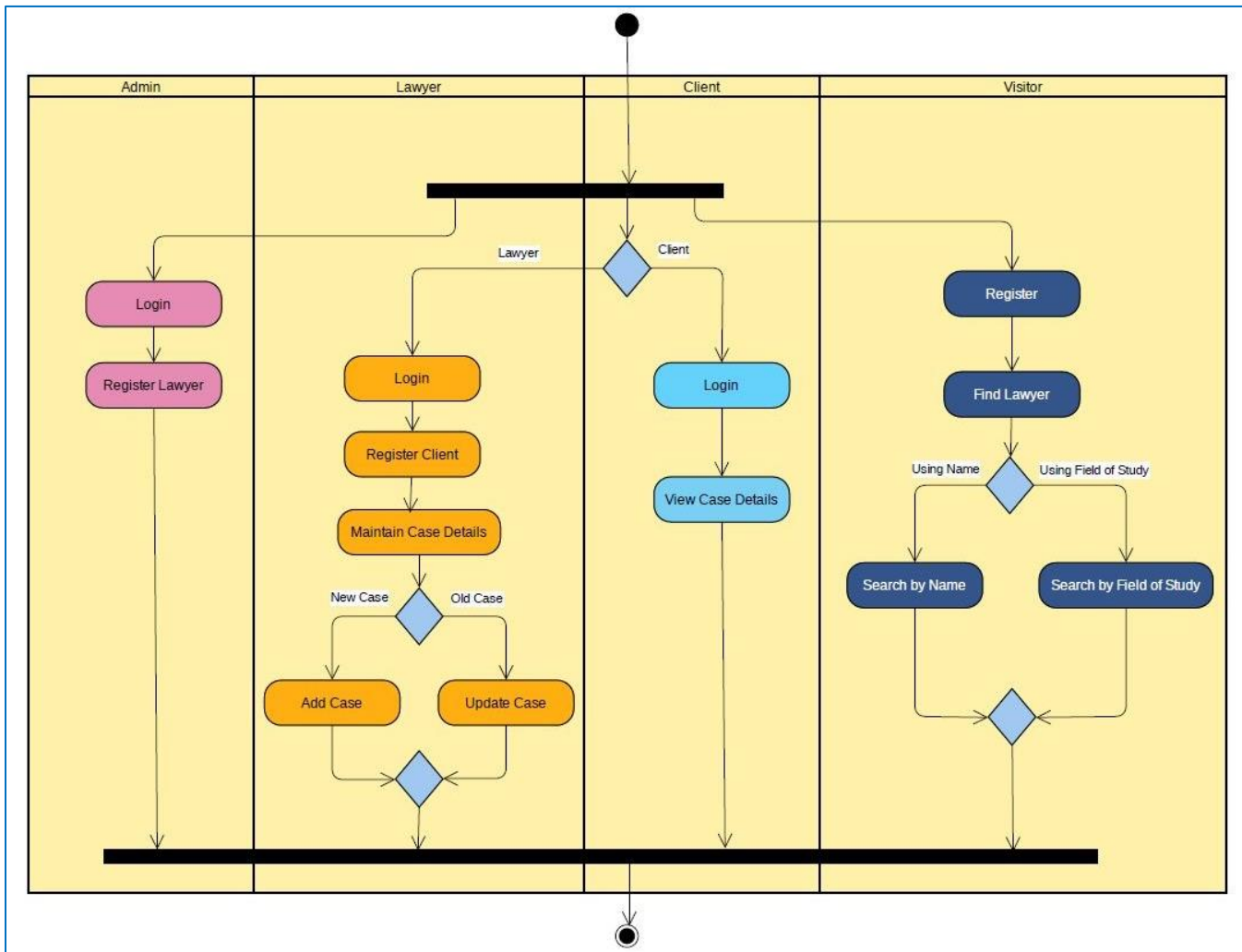
Use-Case Description

An example of a designed Use-Case Description:

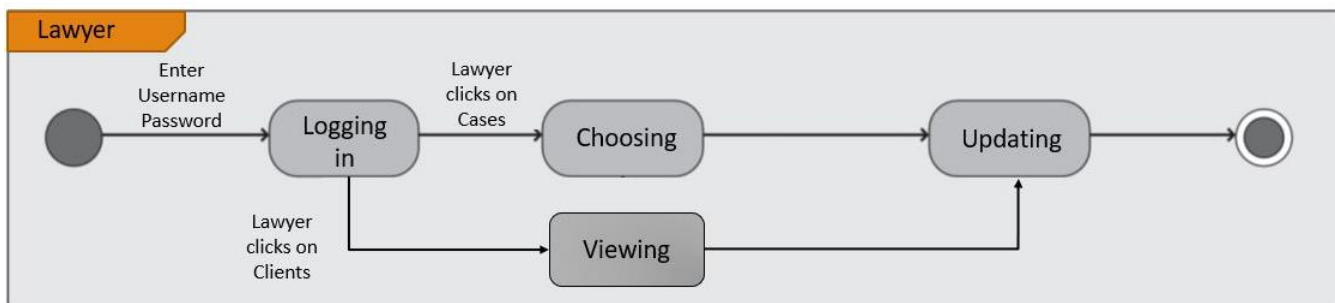
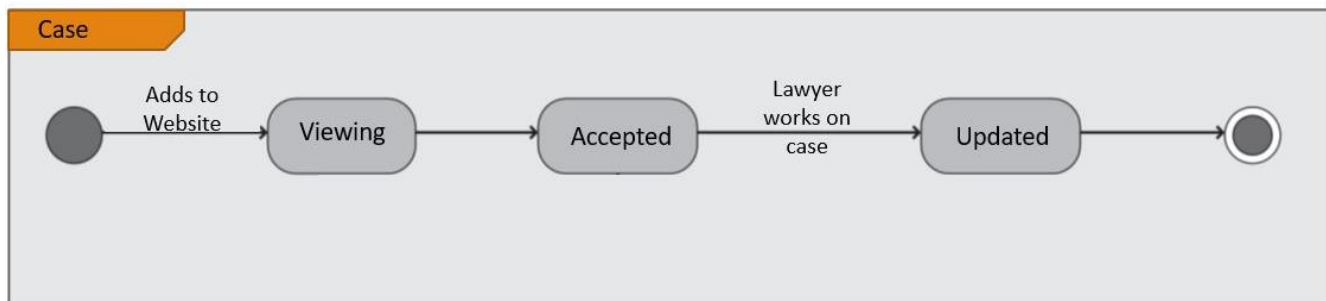
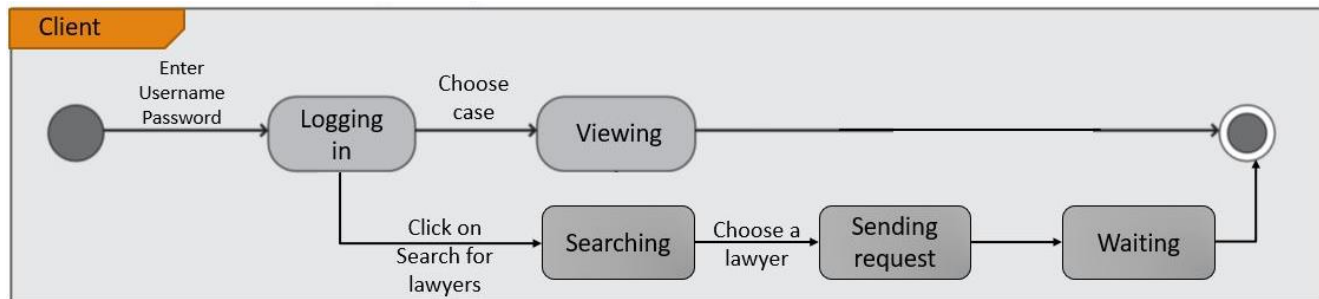
Use Case Name: Add case details	ID: 1	Importance Level : High
Primary Actor: Lawyer	Use Case Type: overview , Essential	
Stakeholders and Interests: Lawyer – wants to append case details of Clients Client – wants to view her/his case		
Brief Description : This use case describes how Lawyer can append case details for each Client.		
Trigger : Lawyer asks to append details of Client's case.		
Relationships: Association : Lawyer Include : Login Extend :		
Normal Flow of Events: 1. Client describes the case for the Lawyer . 2. Lawyer append necessary details of Client's case .		

Note: Other Use-Case Descriptions are available in a separate file titled "Use-Case Description.pdf" in this repository.

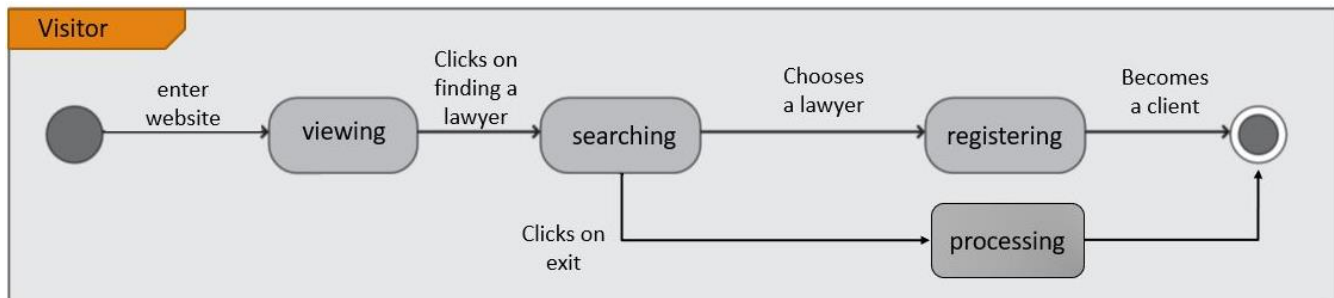
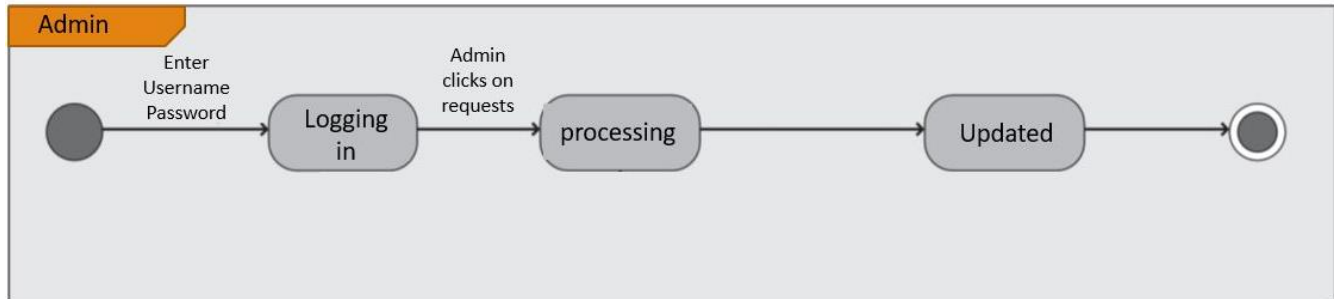
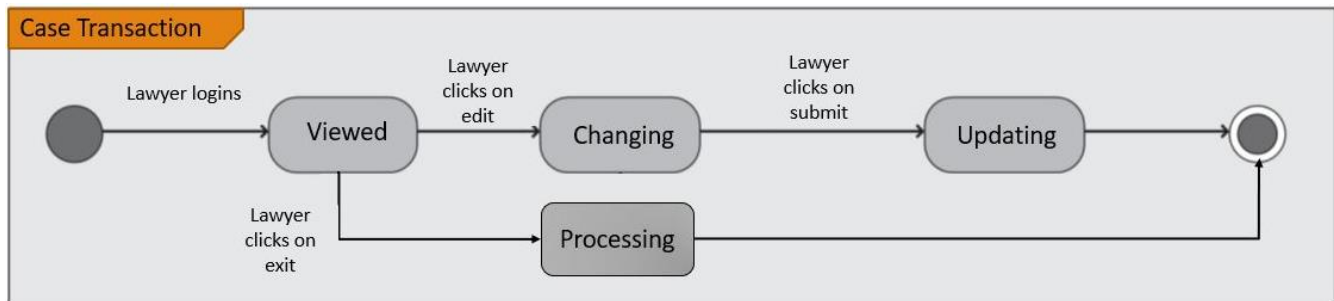
Activity Diagram



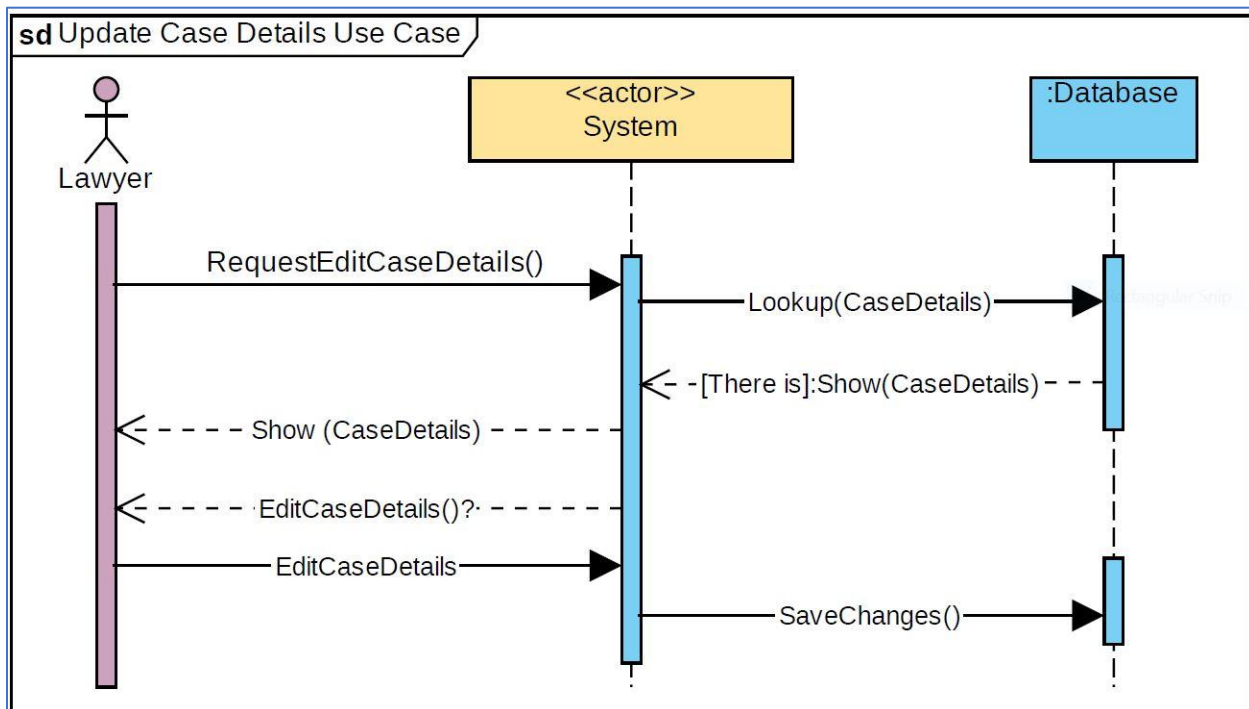
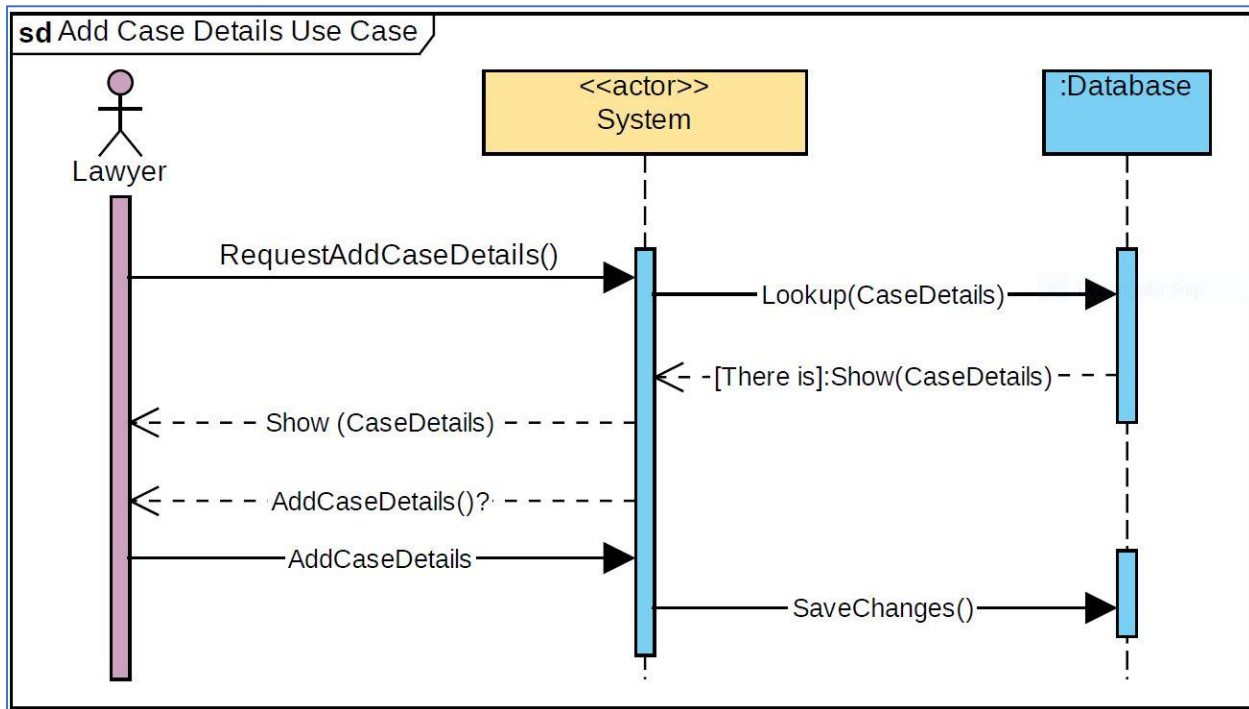
State Machine Diagram (part 1)



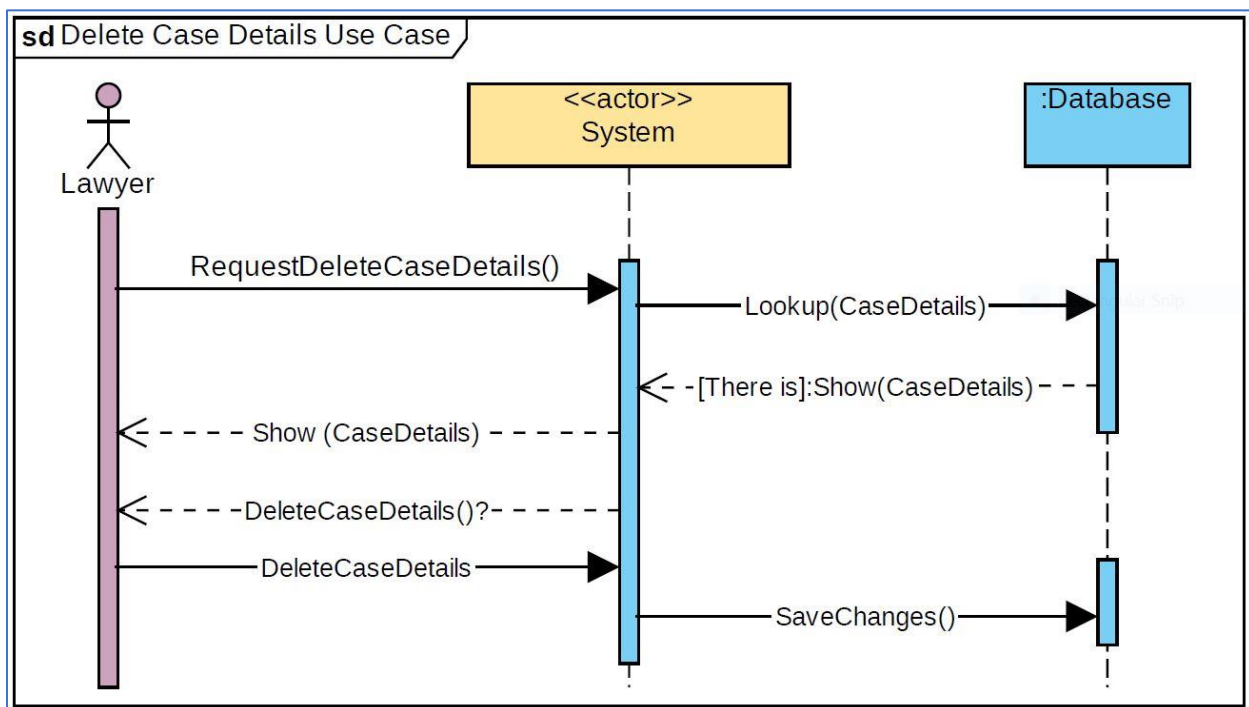
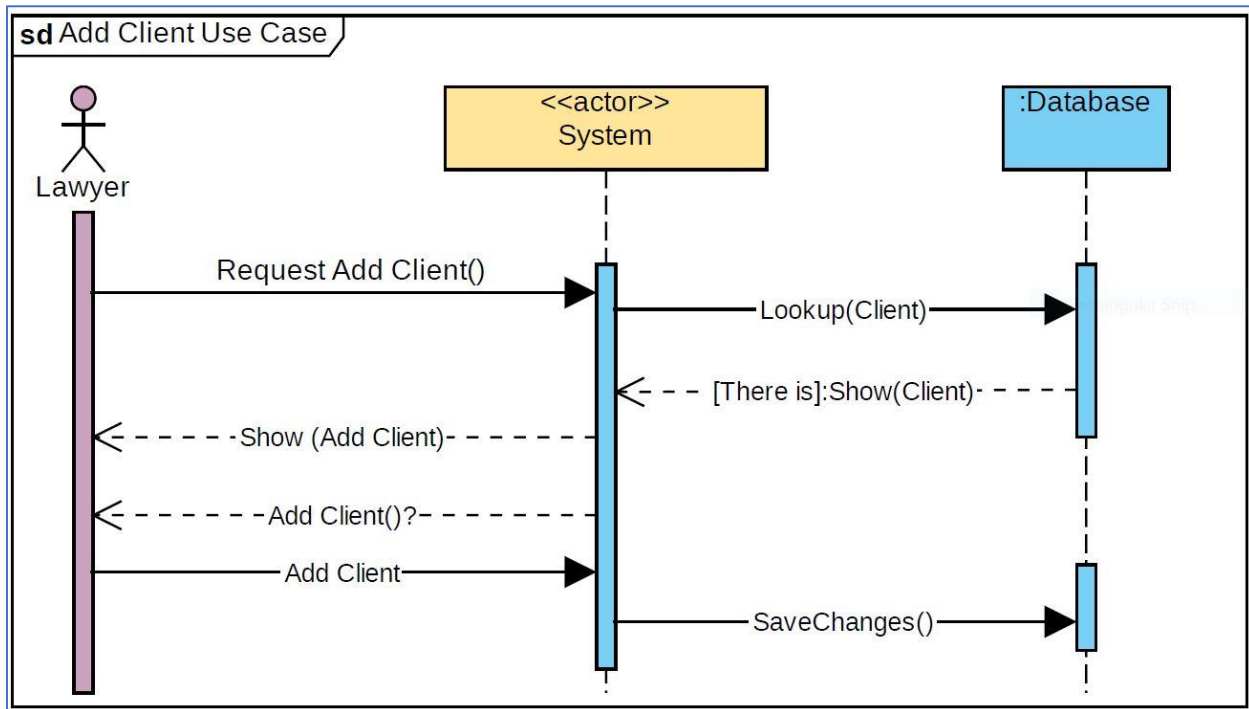
State Machine Diagram (part 2)



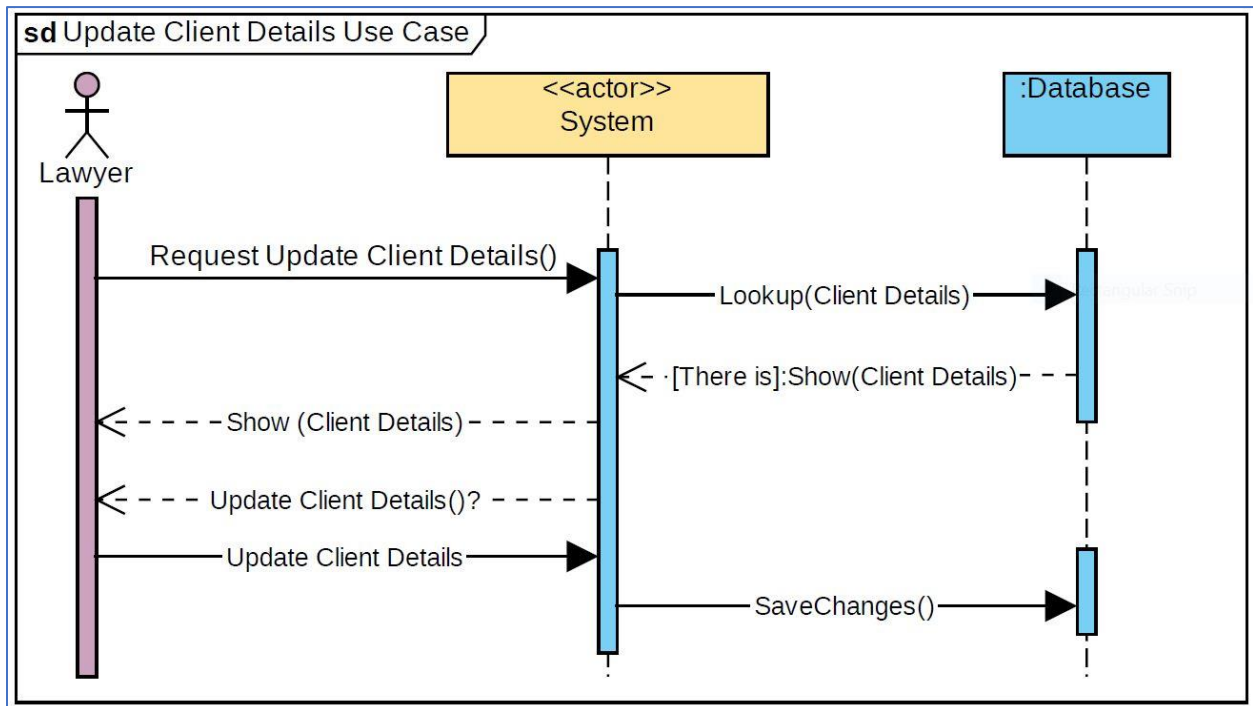
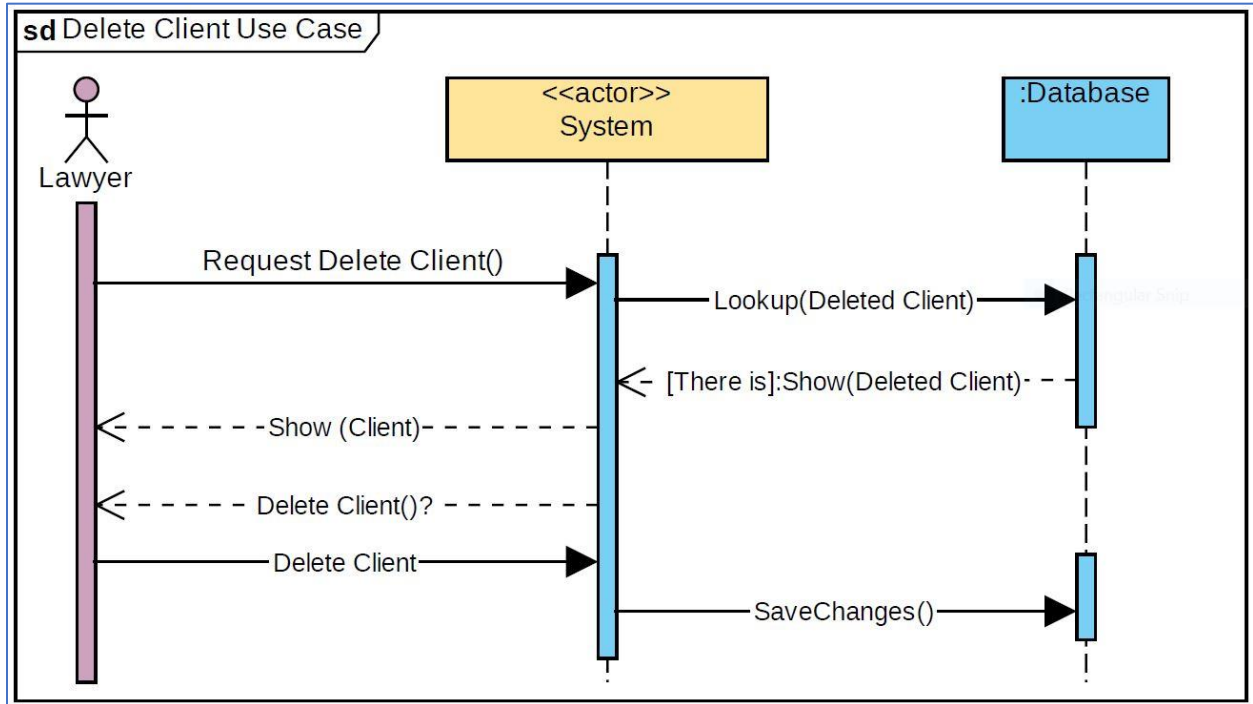
Sequence Diagram(part 1)



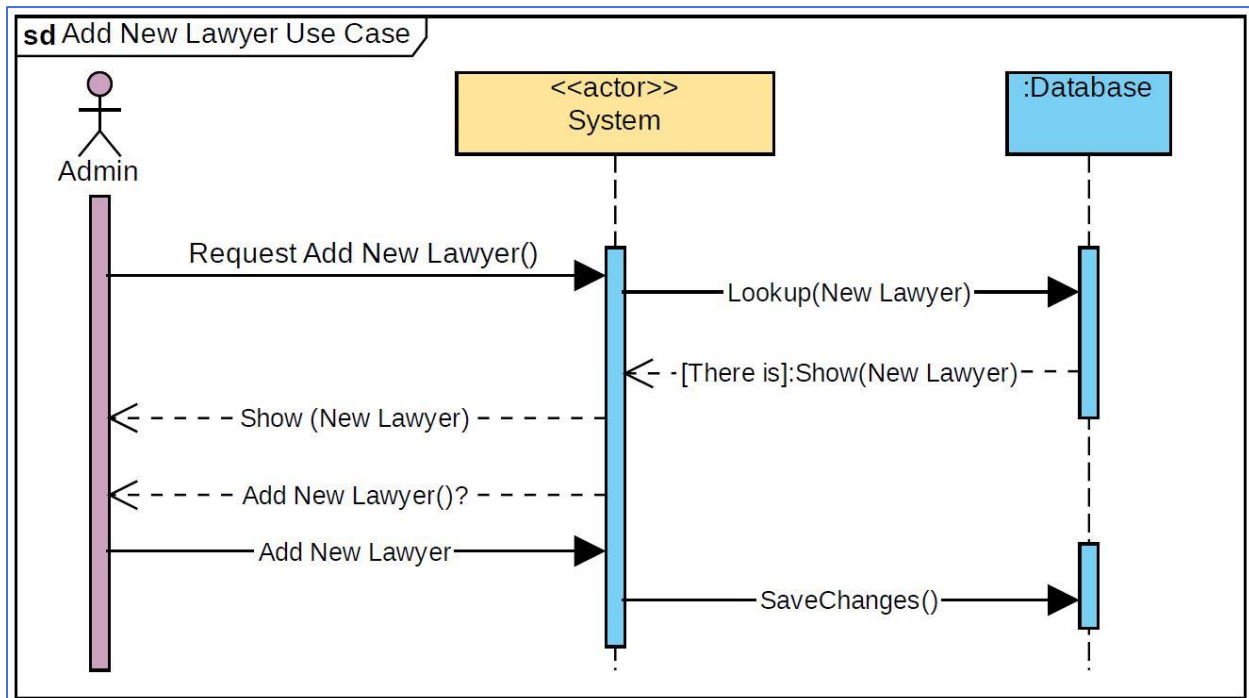
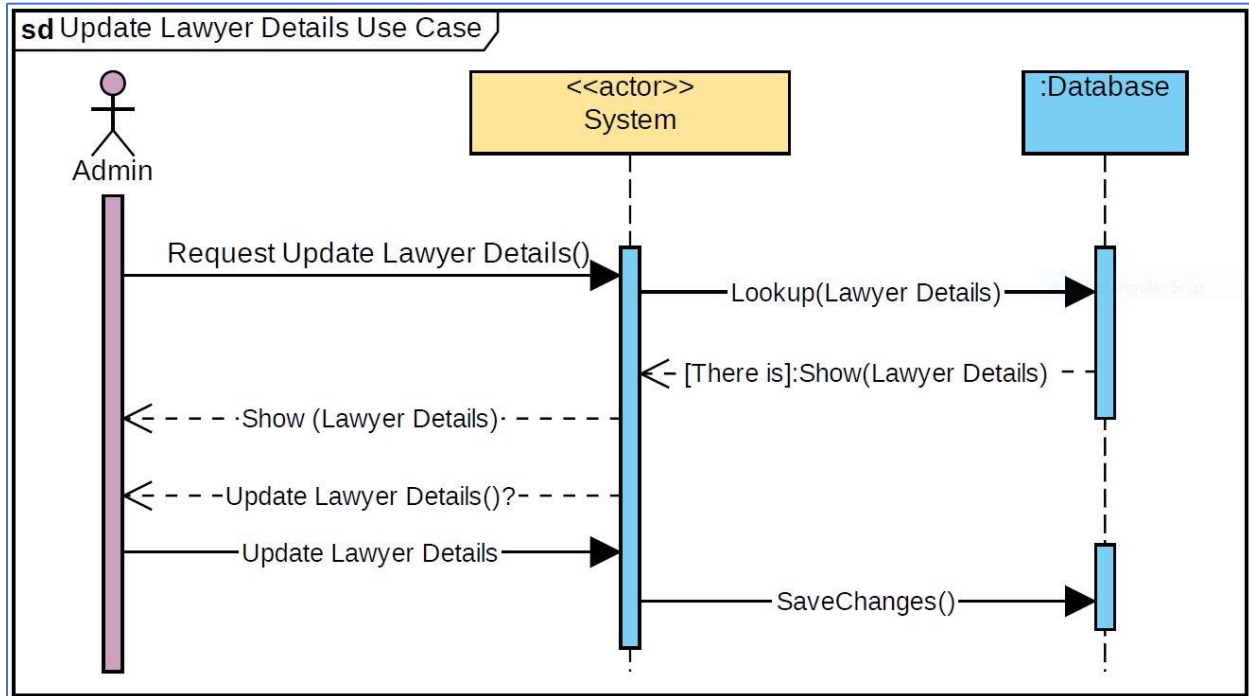
Sequence Diagrams (part 2)



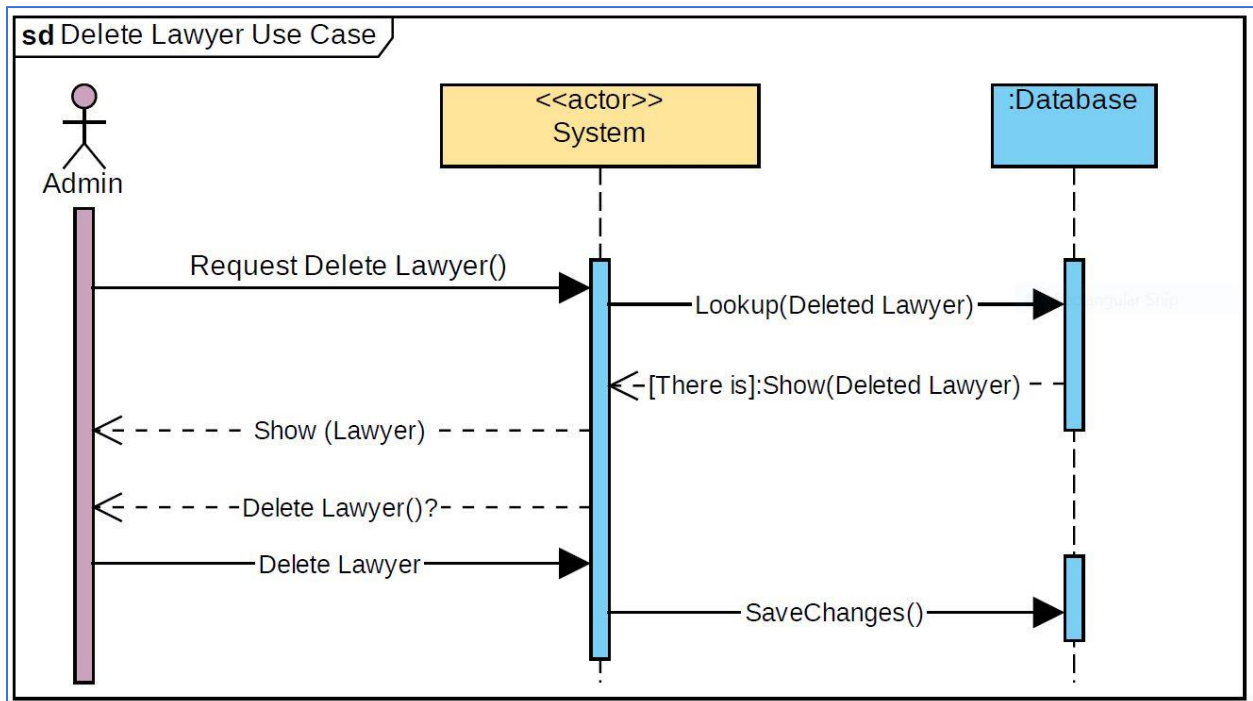
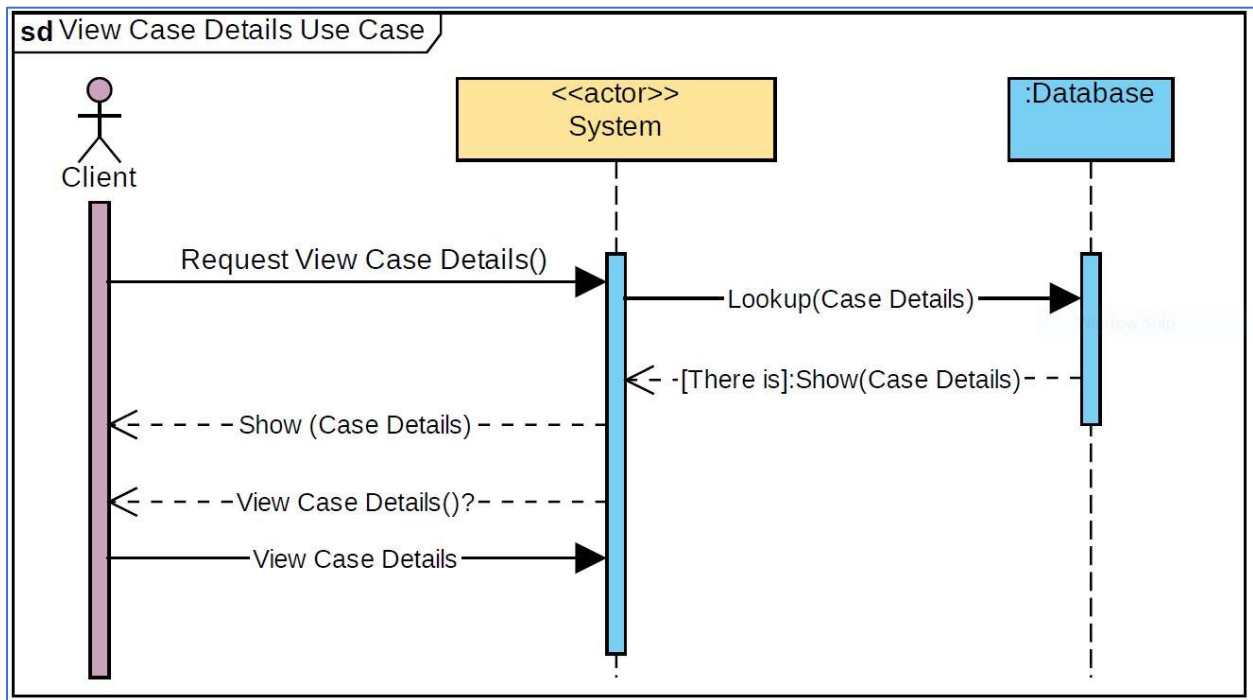
Sequence Diagrams (part 3)



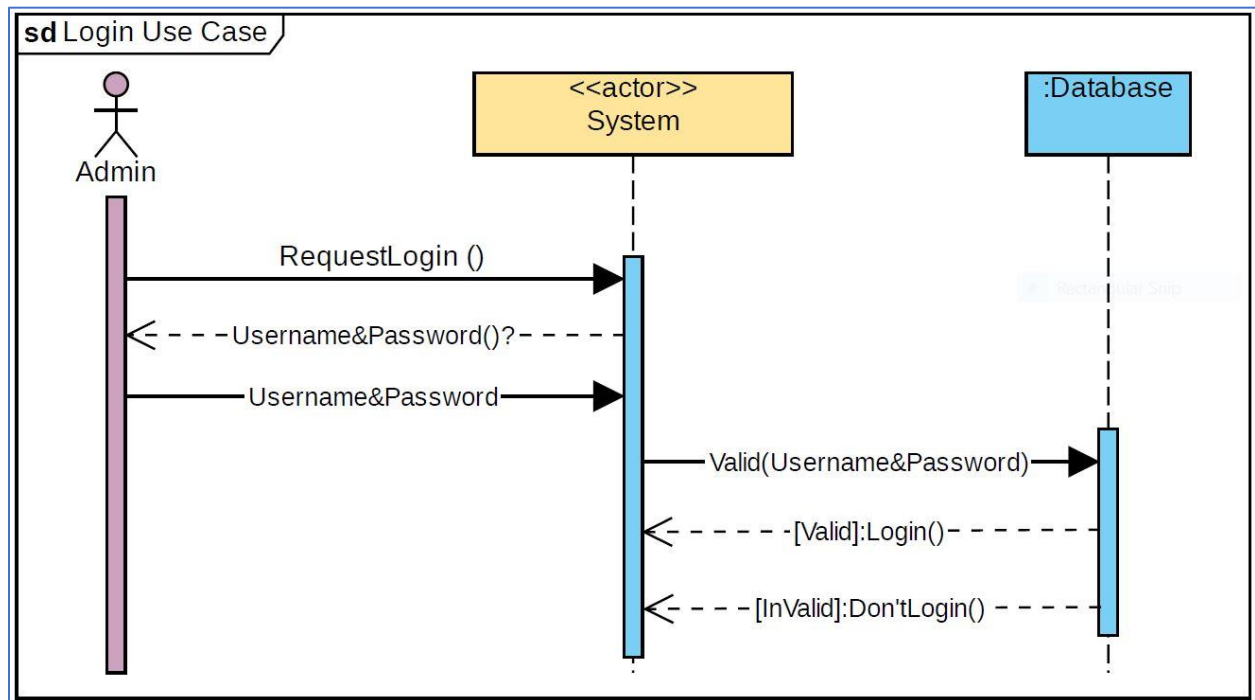
Sequence Diagrams (part 4)



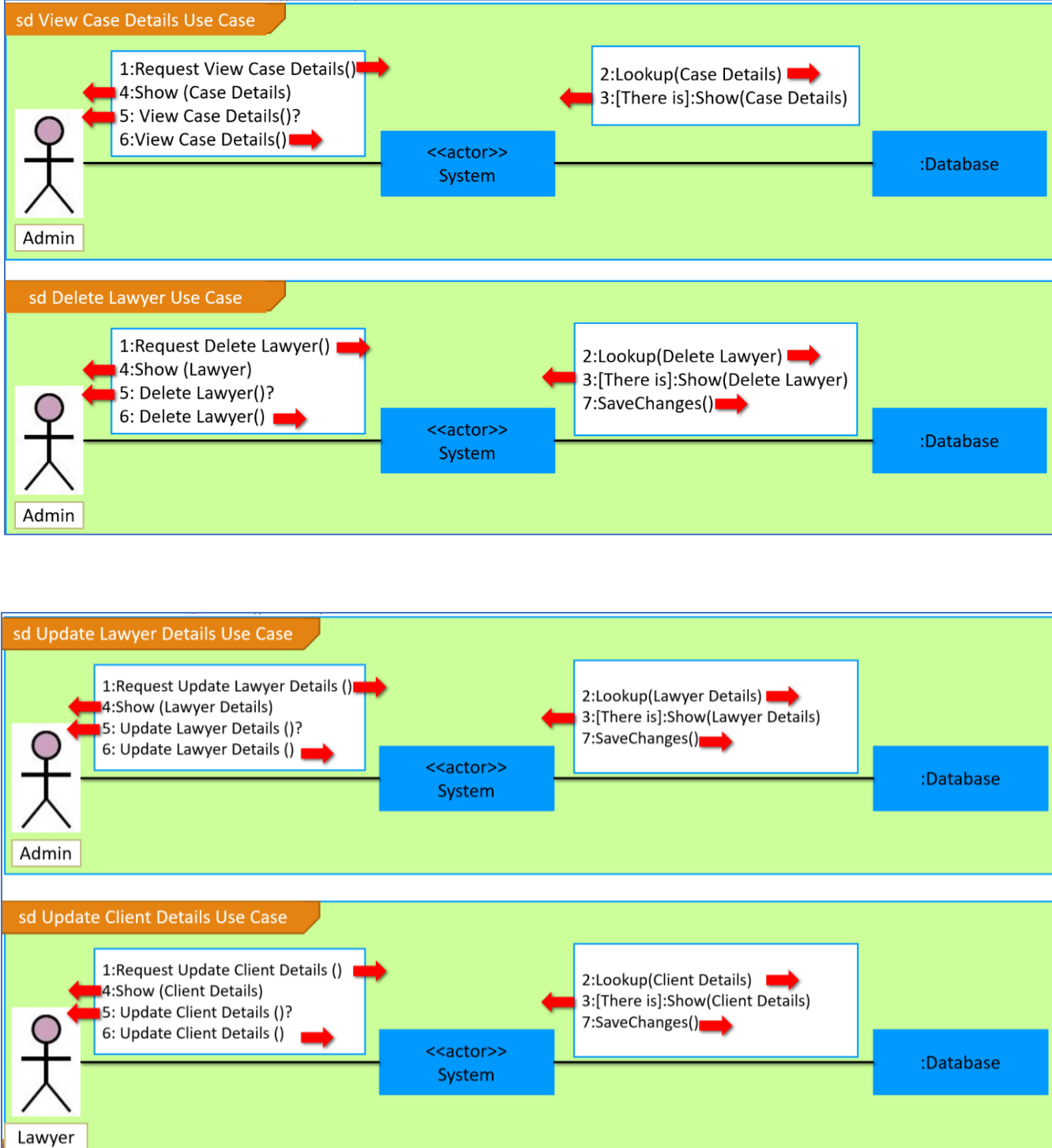
Sequence Diagrams (part 5)



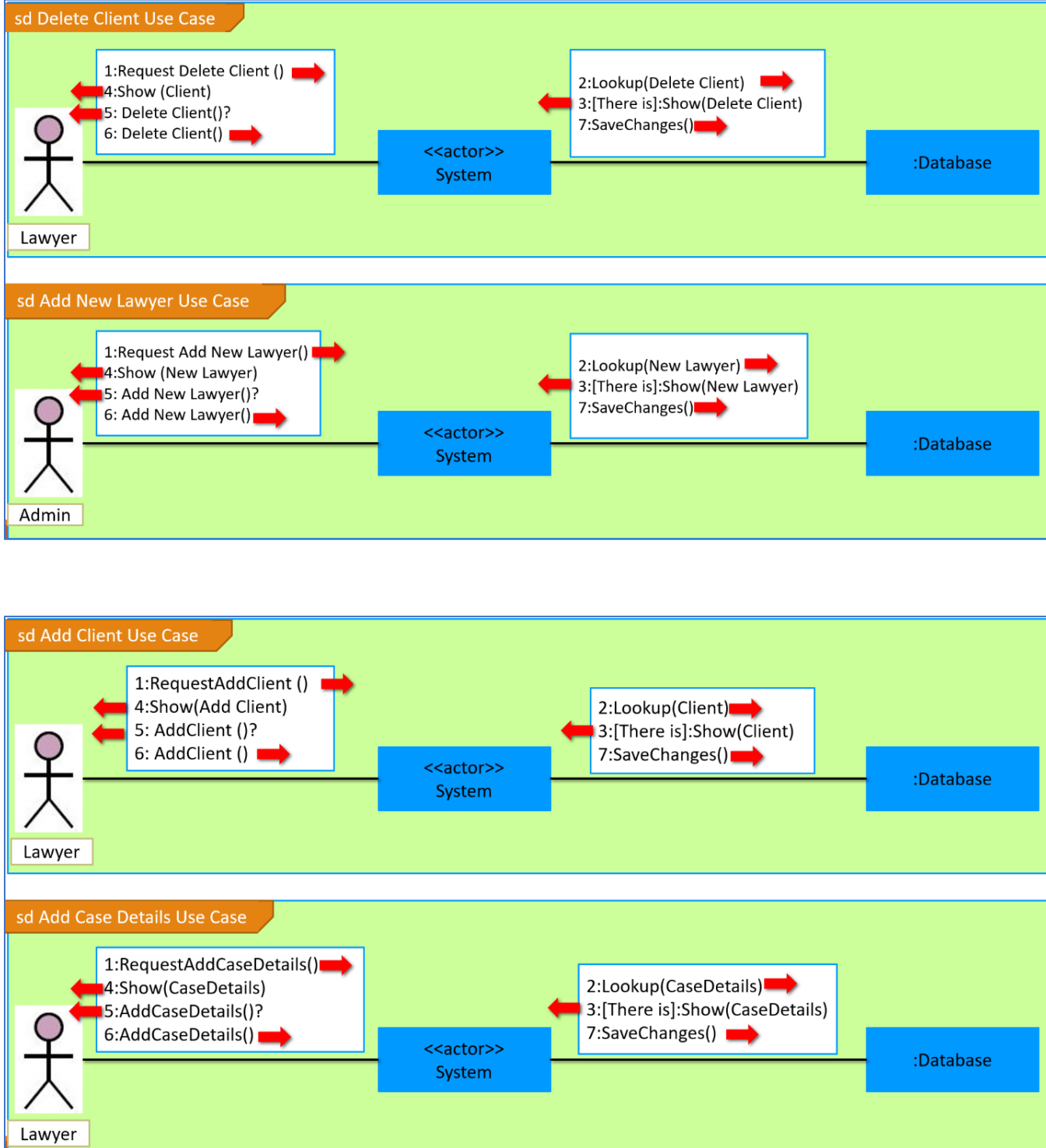
Sequence Diagrams (part 6)



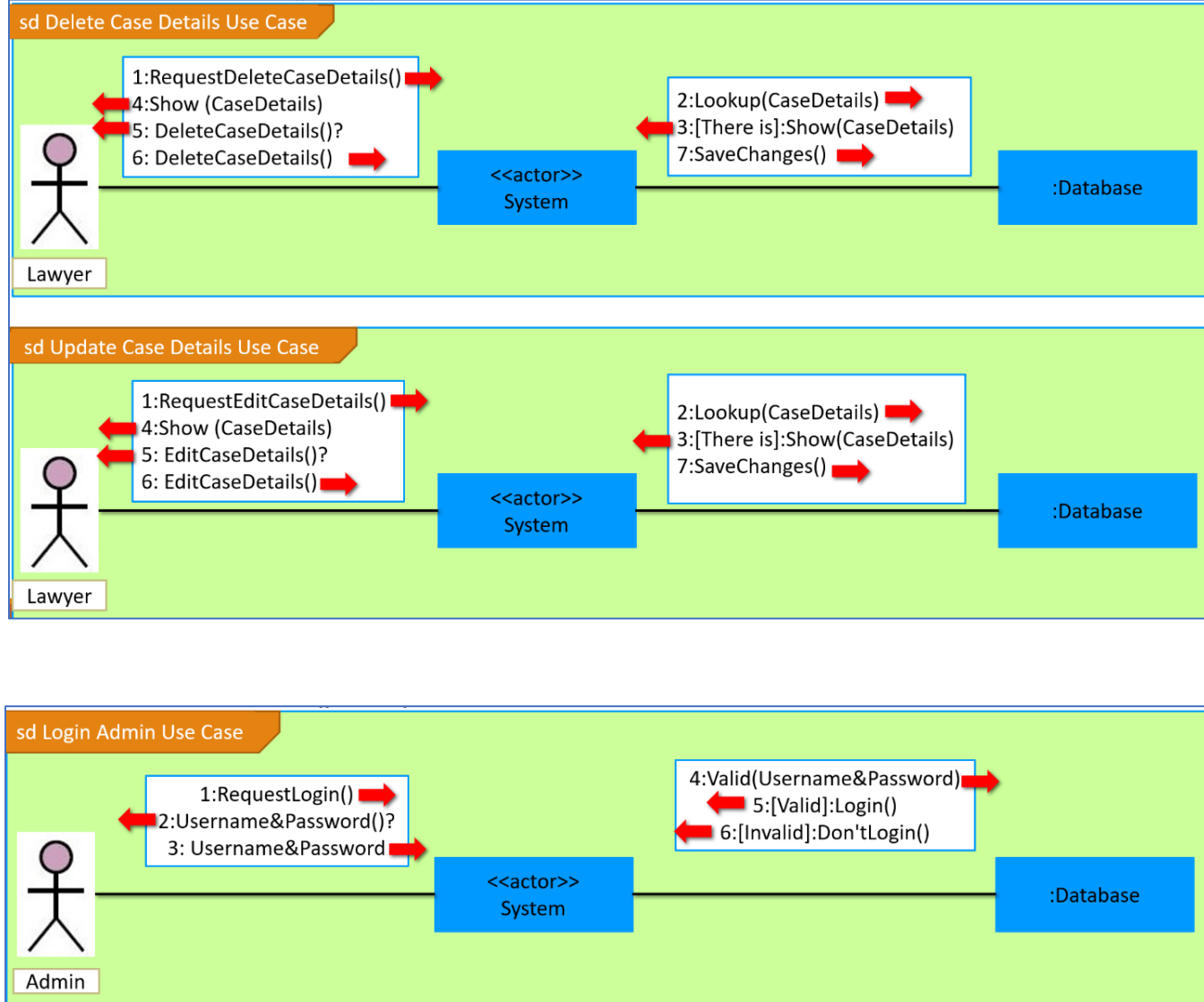
Communication Diagram (part 1)



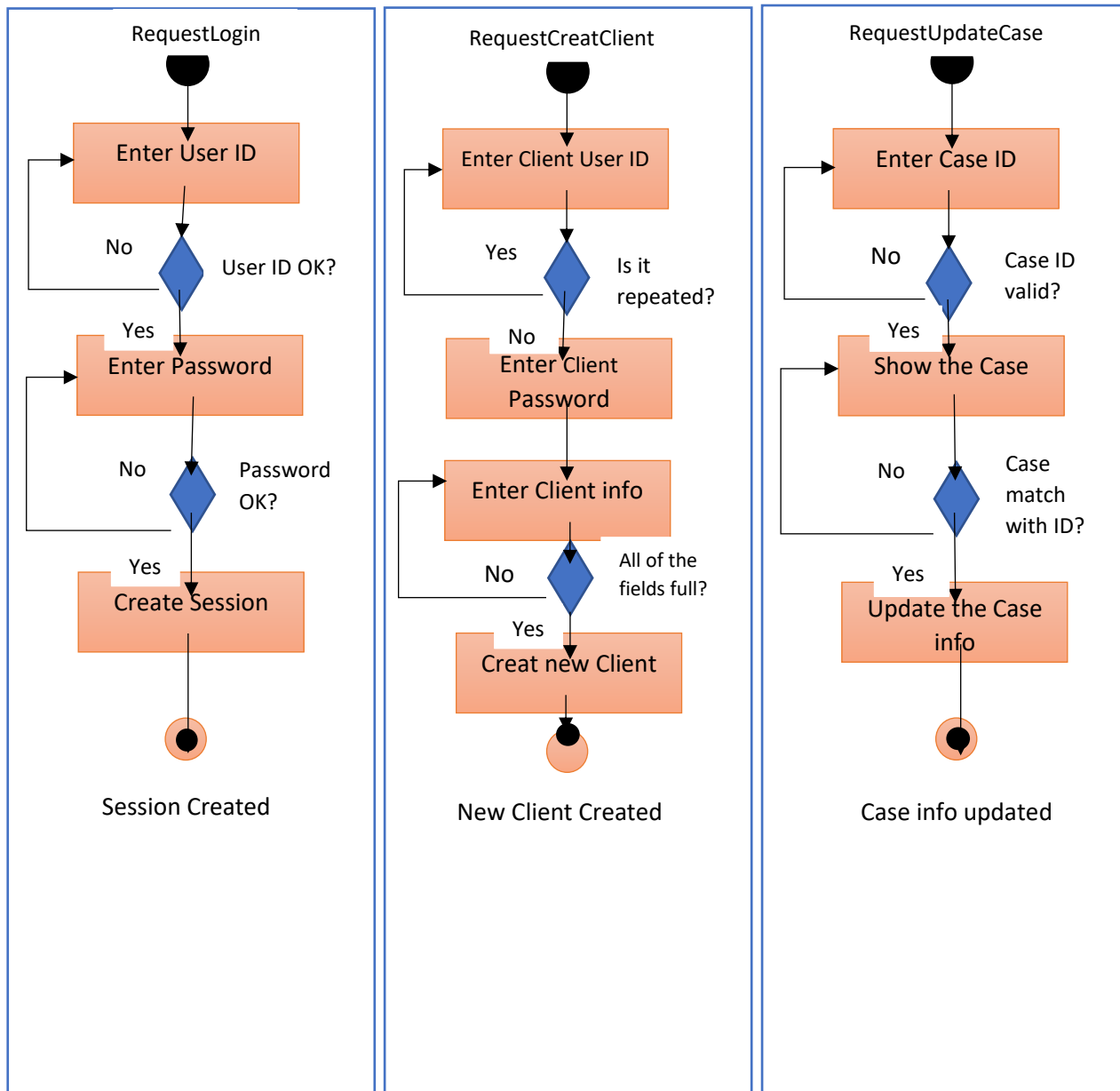
Communication Diagram (part 2)



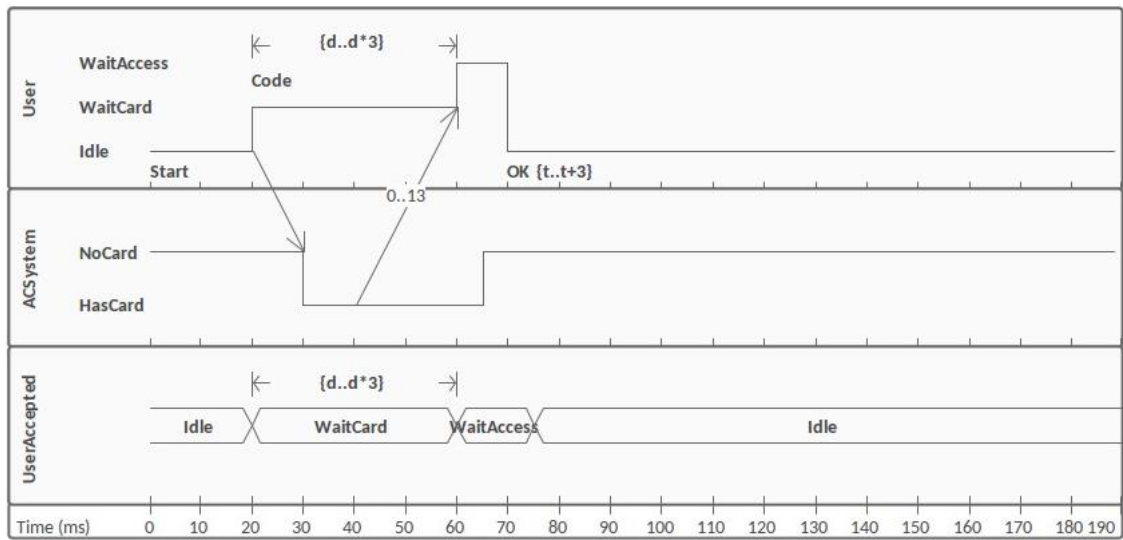
Communication Diagram (part 3)



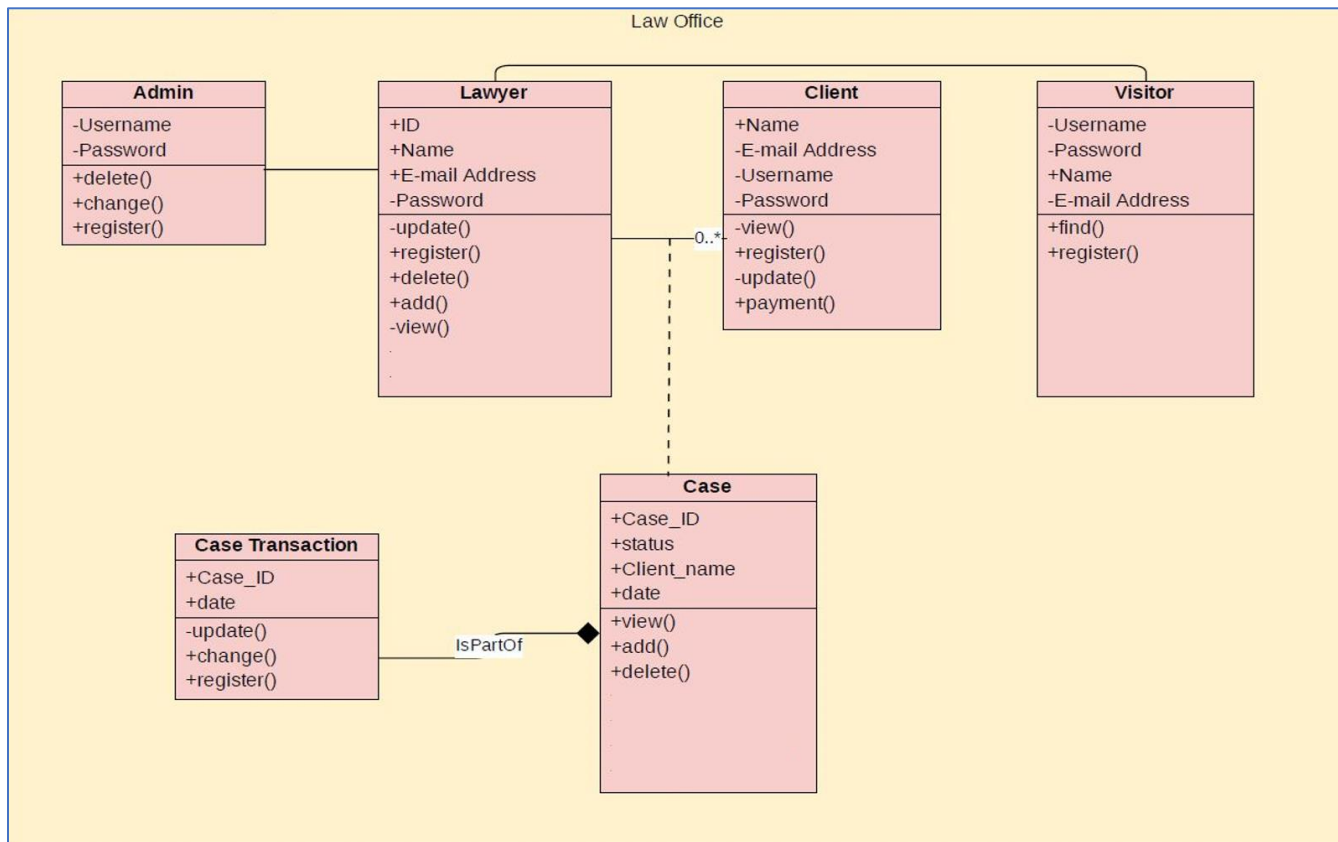
Interaction Overview Diagram



Timing Diagram



Class diagram



CRC card

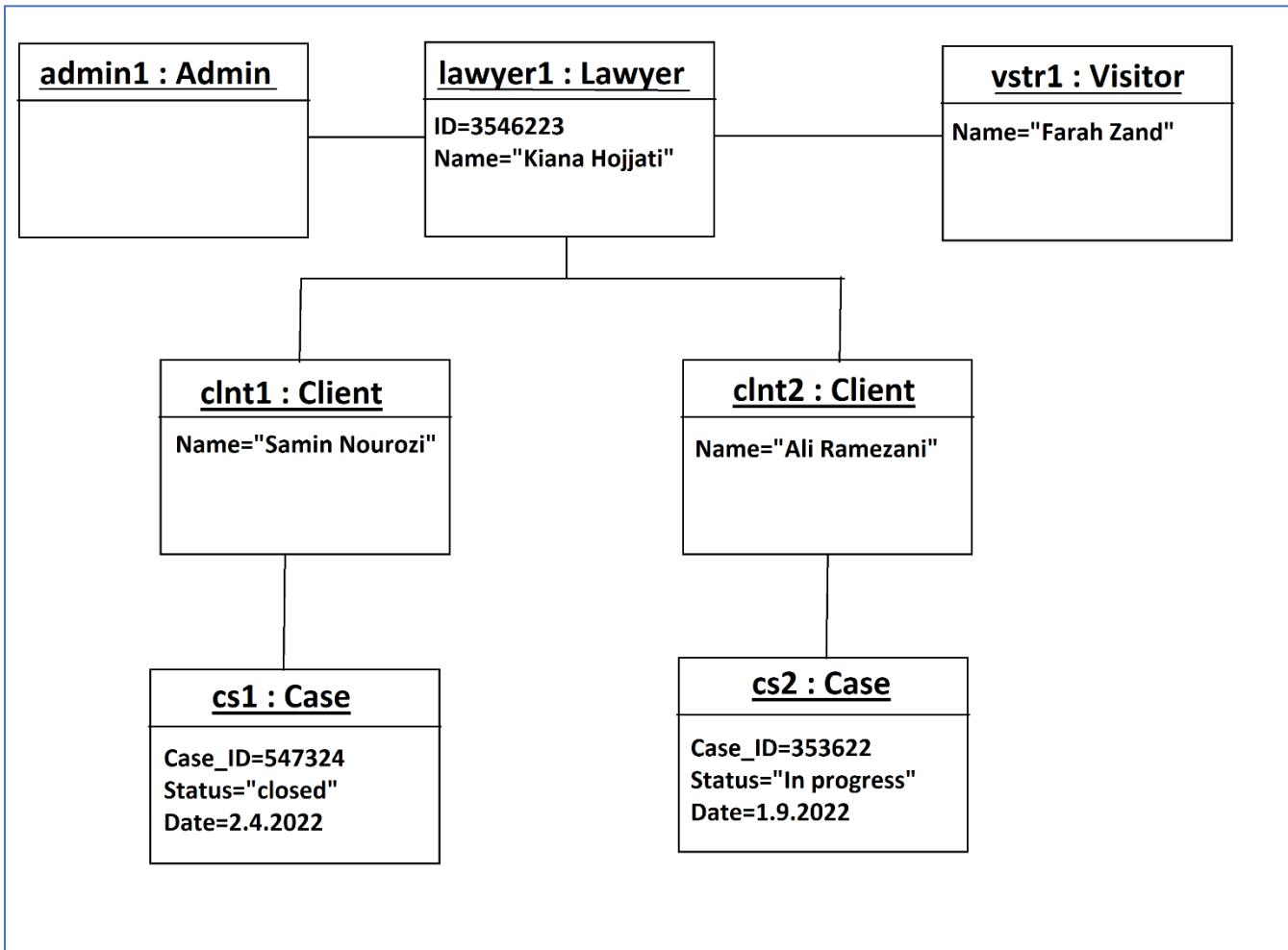
An example of designed CRC cards:

Front:	
Class Name: New Client	ID: 1
Type: Concrete, Domain	
Description: An individual who needs a Lawyer help	Associated Use Cases: 2
Responsibilities Make a new case Change status Save personal information _____ _____ _____ _____ _____	Collaborators Add new Client _____ _____ _____ _____ _____ _____ _____

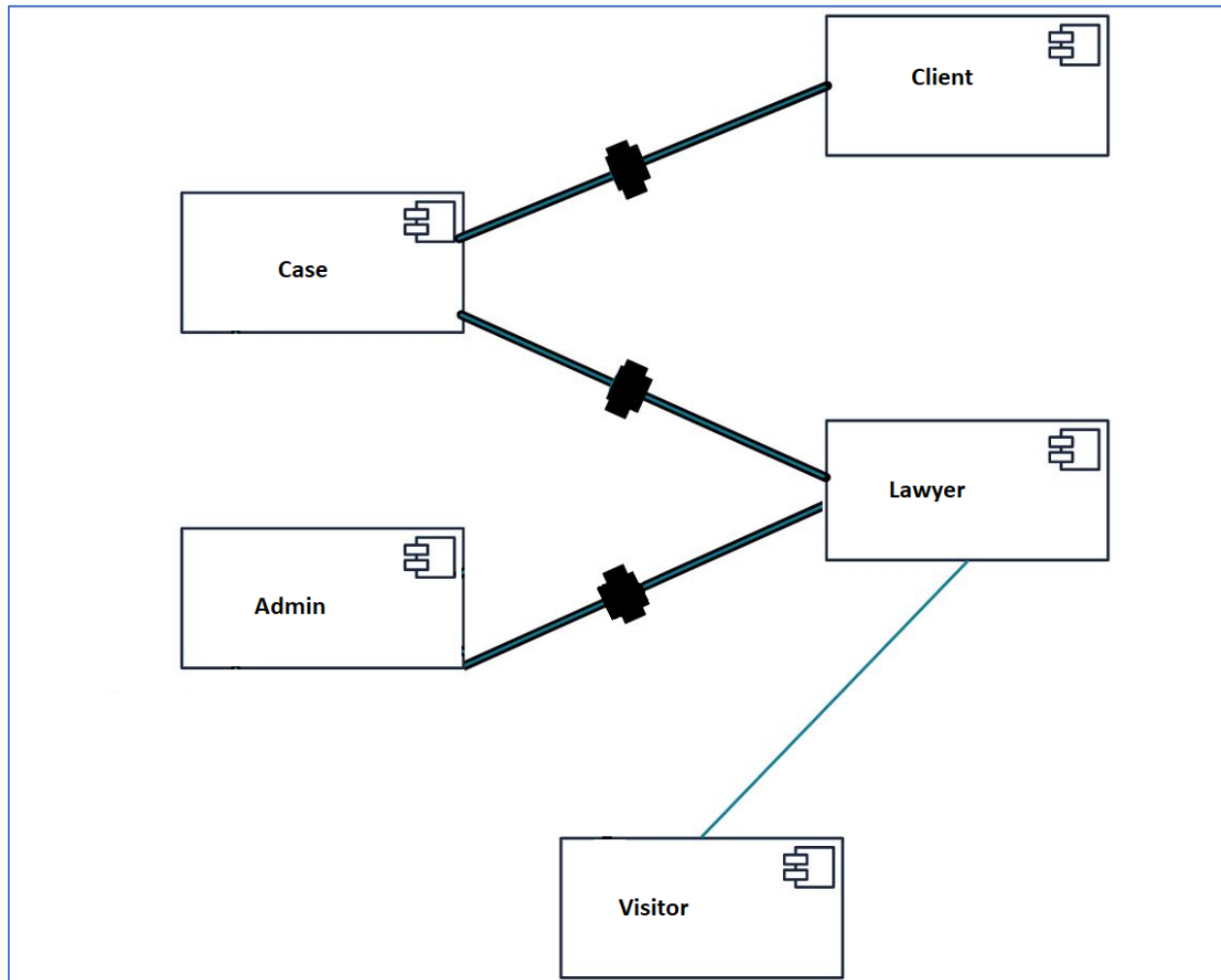
Back:	
Attributes: Amount (double) _____ Insurance carrier (text) _____ _____ _____	
Relationships: Generalization (a-kind-of): Person _____ Aggregation (has-parts): Client history _____ Other Associations: Case , Lawyer _____ _____	

Note: Other CRC cards are available in a separate file called "CRC cards.pdf" in this repository.

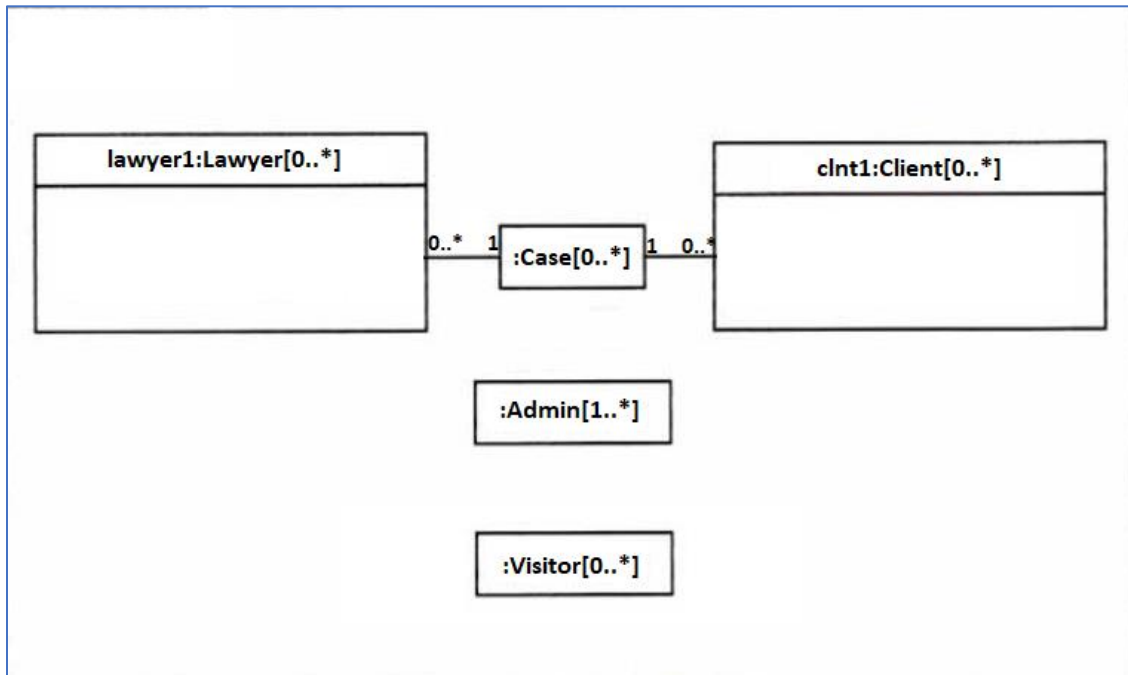
Object Diagram



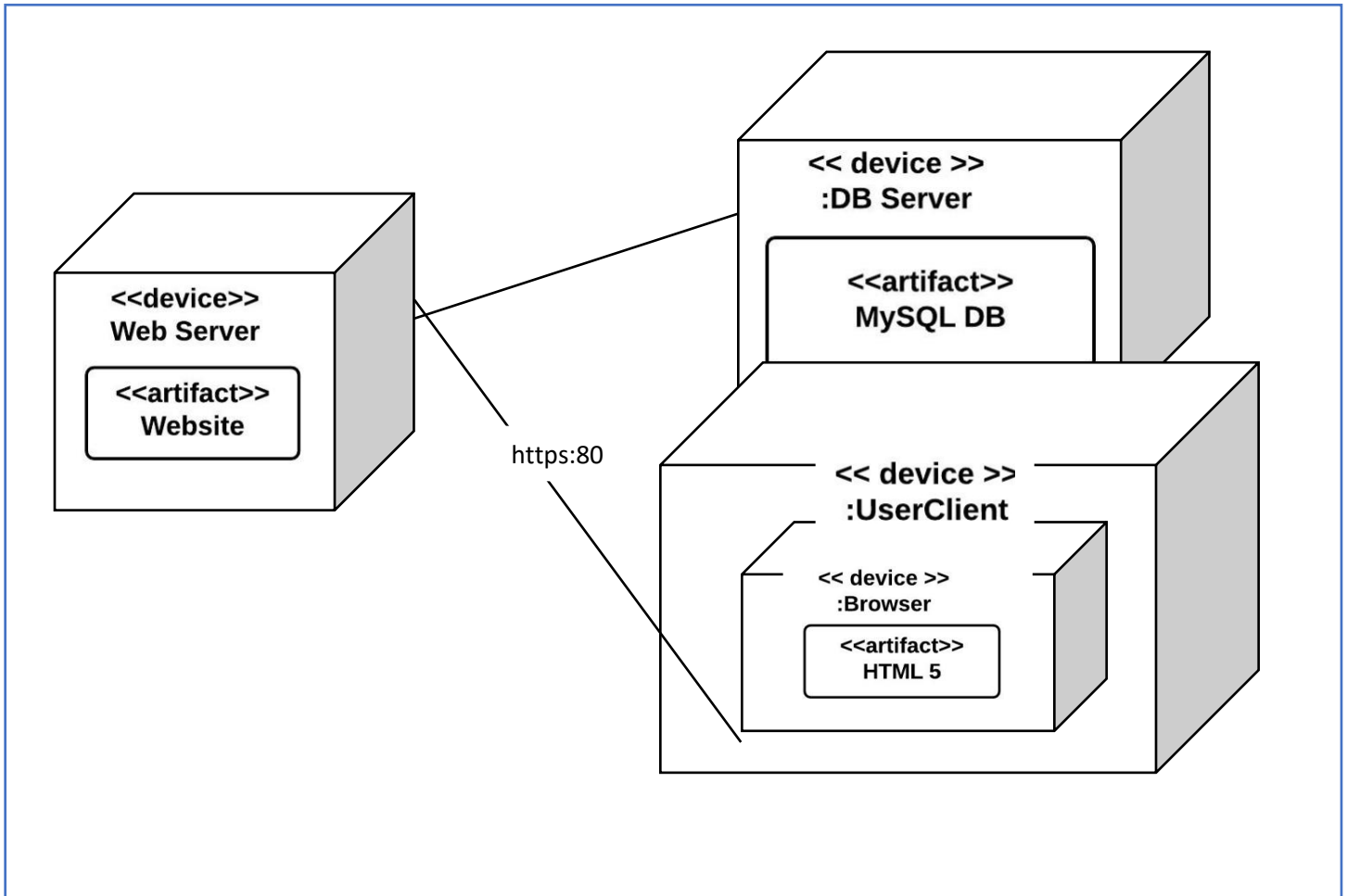
Component Diagram



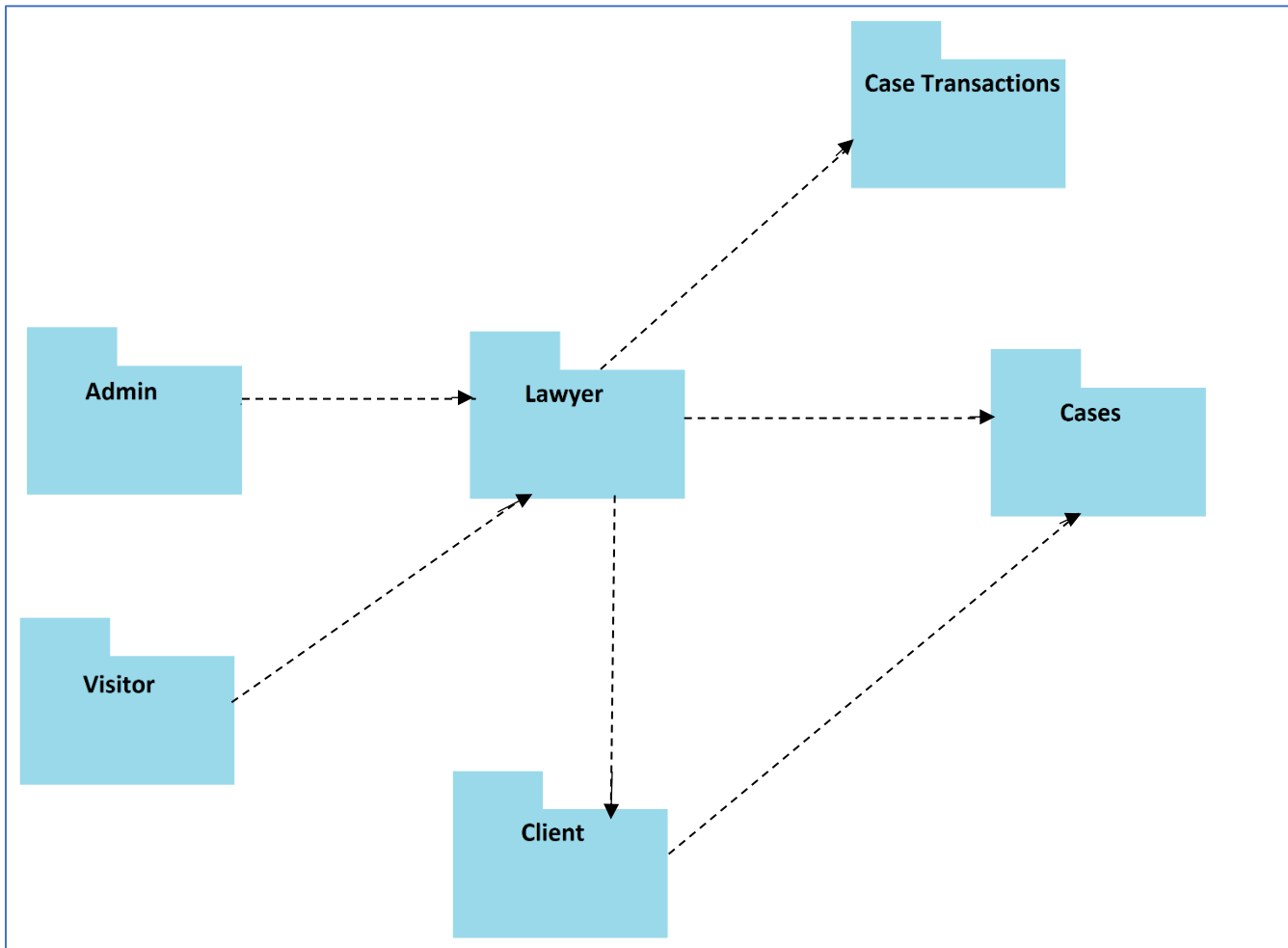
Composite Structure Diagram



Deployment Diagram



Package Diagram



CRUDE ANALYSIS

	Lawyer actor	Admin actor	Client actor	Visitor actor	Admin class	Visitor class	Lawyer class	Client class	Case class	Case transaction class
Lawyer actor			C,D,U				R,E		C,R,U,D, E	R,U,E
Admin actor	C,D,U				R,E					
Client actor								R,E	R	R
Visitor actor						R,E				
Admin class										
Visitor class										
Lawyer class										
Client class										
Case class										
Case transaction class										